ECON 3406: Statistics for Business II (Spring 2027)

General Course Information

Instructor:	Dr. Joshua Martin	**Email:**	joshua@westga.edu
Office:	338 Roy Richards Sr. Hall	**Office Hours:**	MW 1:30-3:00pm, or by apt.
Course Delivery:	In-Person	**Section:**	02
Meetings:	MW 10:30am-11:45am	**Location:**	207 Roy Richards Sr. Hall

Prerequisite

(ECON 3402 or MATH 1401) and MATH 1413

Course Description

This course introduces intermediate quantitative tools for strategic and data-driven decision-making in business. Topics include probability, hypothesis testing, statistical inference, linear regression, multiple regression, logistic regression, and simulation analysis.

In addition to developing a solid foundation in statistical theory, students will gain hands-on experience using *R*, an open-source statistical programming language widely adopted in both industry and research. Emphasis is placed on interpreting statistical results, visualizing and analyzing real-world data, and communicating findings effectively. To support the transition into statistical programming, students will also gain guided exposure to the use of Large Language Models (e.g., "artificial intelligence") as supplementary tools for learning, troubleshooting, and code interpretation.

Course Textbook

David M. Diez, Mine Çetinkaya-Rundel, and Christopher D. Barr. OpenIntro Statistics (Fourth Edition)

Course Goals

The learning goals for this course are as follows:

1. Recognize the importance of data collection, identify limitations in data collection methods and other sources of statistical bias, and determine their implications and how they affect the scope of inference.
2. Use statistical software to summarize data numerically and visually, and to perform data analysis.
3. Have a conceptual understanding of the unified nature of statistical inference.
4. Apply estimation and testing methods to analyze single variables or the relationship between two variables in order to understand natural phenomena and make data-based decisions.
5. Model numerical response variables using a single explanatory variable or multiple explanatory variables in order to investigate relationships between variables.
6. Interpret results correctly, effectively, and in context without relying on statistical jargon.
7. Critique data-based claims and evaluate data-based decisions.

Technology Requirements

You will need to have access to a personal computer or tablet with reliable access to the internet for daily in-person class usage and for the completion of coding exercises.

Technology Support and Guidance Available for Students: TechWest Student ITS Support; UWG Online; 24/7 CourseDen (Brightspace D2L) support

Grading

Evaluation Policies

- All exams are closed notes and must be completed in class.
  - Any substantiated or suspected academic dishonesty will result in immediate course failure and referral for further disciplinary action.
- Make-up Policy:
  - There are no make-ups for missed assignments.
  - Assignments missed due to an excused absence will be replaced with the average of the remaining assignment scores.
  - Excused absences require documentation from a medical professional or relevant university official.
  - Two or more missed, excused exams will result in an incomplete grade for the semester.
- Attendance
  - Most classes will open with a 5 question, 10 minute quiz that is administered online.
  - Completion of these assignments will be used to determine attendance; thus, the grades for the quizzes serve only to signal where focus one's studies before an exam.
  - Students will be randomized into tables before each class and can work together with others at their table.
  - Students must display their (provided) name tent during the quiz in order to receive credit for being present.
  - Non-supervised completed quizzes will be treated as an academic dishonesty incident.

Exams

$$Exams = \begin{cases} 20 \times \sum_{e=1}^{5} Exam_e & \text{if } Exam_5 \leq min(Exam_{1-4}), \\ 20 \times \left(\sum_{e=1}^{4} Exam_e - \min(Exam_e) \right) + 40 \times Exam_5 & \text{if } Exam_5 > min(Exam_{1-4}) \end{cases}$$

where $0 \leq Exam \leq 1$.

Exam Curves

- I write exams to be challenging, so I apply a consistent curve to keep the grading scale fair and to avoid "extra credit–style" outcomes where top scores vastly exceed 100. The size of each student's curve is proportional to their attendance: students who attend every class receive the full curve, and students who never attend receive none.
  - Let g_i denote student i's raw exam score out of 100 points, and let $a_i \in [0, 1]$ denote the share of class meetings student i attended.
  - Each student receives a bonus of $c_i = k \cdot a_i$ points, where $k \geq 0$ is the smallest integer such that either the class mean curved score satisfies $\bar{f} \geq 75$ or the average curve satisfies $\bar{f} - \bar{g} \geq 25$, whichever comes first.
  - For example, if $k = 30$, a student who attended 80% of classes receives $c_i = 0.8 \times 30 = 24$ bonus points, while a student who attended 50% receives $c_i = 15$.
  - Each student's curved score is then:

$$f(g_i) = \min \left\{ \frac{c_i}{2} + \frac{g_i + c_i}{100} \times \left(100 - \frac{c_i}{2} \right),\ 100 \right\}$$

Grade Scale

Grade	Range
A	[90,100+)
B	[80, 90)
C	[70, 80)
D	[60, 70)
F	[0, 60)

Roster Verification

In order to be counted as having attended class during the roster verification period, you must complete at least one quiz by February 1, 2027. If I mark you as not-attending for a course, you will receive a 'W' grade.

Changes to the Syllabus

I reserve the right to modify the course schedule or content as necessary to support the learning objectives of the class. Any such changes will be communicated in advance and will only be made in ways that are beneficial to students, such as reducing the scope of material covered or delaying assignment due dates.

Artificial Intelligence Use Policy

Use of Gen AI tools is not permitted for any graded assessments in this course. Submitting work that is entirely or largely generated by AI will be treated as plagiarism and handled in accordance with the academic and disciplinary policies outlined in the UWG Honor Code (see: Student Handbook).

That said, Gen AI tools – such as OpenAI's ChatGPT – will be incorporated in class activities to support learning. For example, I may use AI to generate commented versions of code demonstrated in class, explaining each function and its purpose. This use is intended to help you focus on understanding statistical theory while still gaining confidence with coding concepts and syntax.

Student Etiquette

- Expectation of Course Attendance
  - Students are expected to attend class regularly, remain actively engaged, take notes, and participate in collaborative discussions with their peers.
  - Absences will only be excused with appropriate documentation from a medical professional or a relevant UWG employee.
- Electronic Devices
  - Students may use laptops and tablets to aid in note-taking.
  - No headphones during class time.
  - Students who need to text or make a call should step outside the classroom.
  - Lecture slides will be provided to students in advance, and in return, students are not permitted to record or photograph during lectures.
- Communication
  - Face-to-face communication is strongly preferred to email correspondence.
  - I request that non-urgent questions be reserved to times in the classroom or during office hours.
  - Students who email questions already answered in the syllabus (including extension requests) will lose points on their next exam:
    - 1 point (out of 20) for the first offense
    - 10 points for the second
    - 20 points on the next exam for the third
- Any disruption to class will result in a zero on the next exam. Any second distribution will result in course failure.

Course Schedule

Date	Readings	Material	Due
01/11/27	Syllabus	Course introduction	
01/13/27	Ch. 1.1	Intro to R	Quiz 1
01/18/27	*MLK Day*		
01/20/27	Ch. 1.2-1.3; CN1	Data fundamentals	Quiz 2
01/25/27	Ch. 1.4; CN2	Experimental design	Quiz 3; Rlab 1
01/27/27	Ch. 2.1	Data generating process	Quiz 4
02/01/27	Ch. 2.2; CN3	Variance and distributions	Quiz 5
02/03/27	Ch. 2.3; CN4	Outliers	Quiz 6; Rlab 2
02/08/27		Ch. 1-2	**Exam 1**
02/10/27	Ch. 3.1	Defining probability	Quiz 7
02/15/27	Ch. 3.2; CN5	Conditional probability	Quiz 8; Rlab 3
02/17/27	Ch. 4.1; CN6	Point estimates	Quiz 9
02/22/27	Ch. 4.2-4.3	Standard errors	Quiz 10; Rlab 4
02/24/27	Ch. 5.1; CN7	Confidence intervals	Quiz 11
03/01/27	Ch. 5.2-5.3; CN8	Hypothesis testing	Quiz 12; Rlab 5
03/03/27		Ch. 3-5	**Exam 2**
03/08/27	Ch. 6.1-6.2; CN9	Intro to statistical inference	Quiz 13
03/10/27	Ch. 6.3; CN10	Inference with proportions	Quiz 14
03/15/27	*Spring Break*		
03/17/27	*Spring Break*		
03/22/27	Ch. 6.4; CN11-12	Chi-squared tests	Quiz 15; Rlab 6
03/24/27	Ch. 7.1-7.2; CN 13-14	T-statistics	Quiz 16
03/29/27	Ch. 7.3-7.4; CN 15-16	Power analyses	Quiz 17
03/31/27	Ch. 7.5; CN17-18	Analysis of variance (ANOVA)	Quiz 18; Rlab 7
04/05/27		Ch. 6-7	**Exam 3**
04/07/27	Ch. 8.1	Correlation	Quiz 19
04/12/27	Ch. 8.2	Least squares regression	Quiz 20
04/14/27	Ch. 8.3-8.4	Regression inference	Quiz 21; Rlab 8
04/19/27	Ch. 9.1-9.2; CN19	Multiple regression	Quiz 22
04/21/27	Ch. 9.3-9.4	Regression diagnostics	Quiz 23
04/26/27	Ch. 9.5	Logistic regression	Quiz 24; Rlab 9
04/28/27	*Flex Day*		Quiz 25
05/03/27		Ch. 8-9	**Exam 4**
05/05/27		Ch. 1-9	**Final Exam**

Student Resources

Accessibility Services

Students with a documented disability may work with UWG Accessibility Services to receive essential services specific to their disability. All entitlements to accommodations are based on documentation and USG Board of Regents standards.

If a student needs course adaptations or accommodations because of a disability or chronic illness, or if they need to make special arrangements in case the building must be evacuated, the student should notify their instructor in writing and provide a copy of their Student Accommodations Report (SAR), which is available only from Accessibility Services. Faculty cannot offer accommodations without timely receipt of the SAR; further, no retroactive accommodations will be given.

For more information, please contact Accessibility Services.

Center for Academic Success

The Center for Academic Success provides services, programs, and opportunities including academic coaching, peer tutoring, drop-in tutoring and supplemental instruction to help all undergraduate students succeed academically.

For more information, contact them: (678) 839-6280 or cas@westga.edu.

Mental Health Support

If you or another student are experiencing a mental health issue, free and confidential services are available through the Counseling Center.

Students who have experienced sexual or domestic violence can access confidential support through Patient Advocates in Health Services.

To report a concern anonymously, please visit UWG Cares.

English Language Learning (ELL) Resources

If you are having difficulty with English or U.S. academic culture, specialized resources are available to help you succeed. Visit the ELL Resource Page for more information.

Title IX & Equal Opportunity

University of West Georgia faculty are committed to helping create a safe and open learning environment for all students. If you (or someone you know) have experienced any form of sexual misconduct, including sexual assault, dating or domestic violence, or stalking, know that help and support are available. The University strongly encourages all members of the community to take action, seek support, and report incidents of sexual misconduct to the Office of Title IX & Equal Opportunity. Please be aware that under Title IX of the Education Amendments of 1972, I am required to disclose information about such misconduct to the Office of Title IX & Equal Opportunity. If you wish to speak to a confidential employee who does not have this reporting responsibility, you can contact the University of West Georgia Counseling Center (678-839-6428) or Advocate Services (470-215-9946).